Multiplication Masters

The mission where multiplication becomes a shape.

THE BIG QUESTION

Why does the area model give the same answer as the standard algorithm?

HANDS-ON PROJECT

Launch Bay Blueprint

Design a rocket hangar on graph paper. Find its area two different ways. Prove they match.

FLUENCY GAME

Product Blockout

Two dice, multiply, shade that rectangle on a 10×10 grid. First to block the board wins.

AXIOM · MISSION COMMANDER, NINE TOURS

"Brock — you're not learning multiplication. You're learning why it works. Different job, much more interesting."

MISSION 01 · BEFORE WE LAUNCH

Pre-Assessment

NAME ______________

DATE ______________

Twelve items, five strands. Anything you already own, we skip.

This is not a test. Nothing here counts for a grade. It exists to delete work you don't need. If you don't know something, write “not yet” and move on — that's a useful answer.

STRAND A FACTS

A1 $7 \times 8 =$ ______

A2 $9 \times 6 =$ ______

STRAND B AREA MODELS

B1 Draw 4 × 13 as a rectangle.

B2 $6 \times 47 =$ ________

STRAND C 2-DIG × 2-DIG

C1 $23 \times 14 =$ _______

C2 $47 \times 32 =$ _______

STRAND D FACTORS

D1 List every factor pair of 24.

D2 Is 51 prime? How do you know?

STRAND E MULTIPLES

E1 First five multiples of 7.

E2 Which of 2, 3, 5, 9 divide 90?

STRAND F REASONING

F1 19 × 6 in your head. Then write how.

F2 $6 \times 4\square = 276$ $\square =$ _____

TEACHER PAGE · DO NOT HAND TO STUDENT

Skip Chart

Score the pre-assessment, then cross weeks off the calendar.

2 / 2 → SKIP
1 / 2 → COMPRESS
0 / 2 → TEACH

STRAND	ITEMS	COVERS	SCORE	IF 2/2, REMOVE
A · Facts	A1 A2	Single-digit fluency	___ / 2	Warm-Up tier drops to 3 items all unit
B · Area models	B1 B2	Week 1 · lessons 1.1–1.2	___ / 2	Days 1.1 and 1.2 — keep Friday enrichment
C · 2-dig × 2-dig	C1 C2	Week 1 · 1.4 and all of Week 2	___ / 2	Week 2 days 2.1–2.3. Keep the Big Question talk.
D · Factors	D1 D2	Week 3	___ / 2	Days 3.1–3.3. Keep 3.4 (square numbers) — it's new.
E · Multiples	E1 E2	Week 4 · lessons 4.1–4.2	___ / 2	Days 4.1–4.2. Never skip the sieve (4.3).
F · Reasoning	F1 F2	Week 1 · 1.3 and Week 5 · 5.1	___ / 2	Nothing. Raise the Challenge tier instead.

Where the saved time goes

Never back into the calendar. Compacted time is spent on the Challenge tier, the Launch Bay Blueprint, or an early start on Mission 02. Write the reallocation here so it doesn't quietly evaporate:

....

....

Revised Mission 01 plan

WEEK 1 ☐ full ☐ compressed ☐ skip

WEEK 2 ☐ full ☐ compressed ☐ skip

WEEK 3 ☐ full ☐ compressed ☐ skip

WEEK 4 ☐ full ☐ compressed ☐ skip

WEEK 5 ☐ full ☐ compressed ☐ skip

NAME ___________

DATE ___________

Arrays Become Area

Build it with blocks, then draw it as a rectangle. Same number, better picture.

Warm-Up

6 ITEMS · 5 MINUTES · FACTS YOU ALREADY OWN

6 × 7	8 × 4	9 × 6	7 × 7	12 × 3	11 × 5
____	____	____	____	____	____

Core

SPLIT THE BIG NUMBER INTO TENS AND ONES. LABEL BOTH ROOMS.

1. 4 × 13 = ________

2. 6 × 14 = ________

3. 3 × 27 = ________

4. 5 × 18 = ________

5. 7 × 16 = ________ Write the two partial products before you add them: ....

Challenge

NOT GRADED. GETTING STUCK HERE IS THE POINT.

C1. A rectangle has area 84. One side is 6. How long is the other side? Show how you found it, not just the answer.

....

C2. Find *every* rectangle with whole-number sides and area 36. How do you know you found them all?

....

....

C3. A rectangle with area 37 has only one possible shape. Why? What's special about 37?

....

AXIOM: Don't erase a wrong rectangle. Cross it out and leave it there — I want to see the route you took.

NAME ___________

DATE ___________

Breaking Apart

8 × 34 is hard. 8 × 30 and 8 × 4 are easy. That's the whole trick.

Warm-Up

5 × 9	8 × 8	7 × 8	6 × 9	4 × 12	3 × 11
____	____	____	____	____	____

WORKED EXAMPLE

8 × 34

8 × 30 240	8×4 32

240 + 32 = **272**

Core

TWO ROOMS EACH. WRITE BOTH PARTIAL PRODUCTS.

1. 8 × 34 .... = ________
2. 6 × 47 .... = ________
3. 9 × 26 .... = ________
4. 7 × 58 .... = ________
5. 4 × 96 .... = ________

Challenge

C1. Break 8 × 34 apart a completely different way — not into tens and ones. Show that you still get 272.

....

C2. Fill in both digits. 6 × 4□ = 2▲6 □ = ____ ▲ = ____

Explain the first thing you noticed that narrowed it down:

....

C3. Is 8 × 34 the same as 4 × 68? Answer without multiplying, then check.

....

Mental Math Moves

NAME ___________

DATE ___________

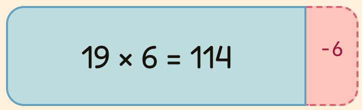
Go past the number, then take back the extra. No pencil for the Core tier today.

✦ Warm-Up FRIENDLY NUMBERS

25 × 4	25 × 8	50 × 6	20 × 7	15 × 4	30 × 9
___	___	___	___	___	___

THE MOVE

$19 \times 6 = 20 \times 6 - 6 = 114$



The dashed strip is the extra column you added and then gave back. Draw it every time until you don't need to.

◆ Core ANSWER, THEN NAME THE MOVE YOU USED

1. 19 × 6 ....
2. 99 × 7 ....
3. 48 × 5 ....
4. 25 × 12 ....
5. 102 × 4 ....
6. 35 × 6 ....

★ Challenge

C1. Draw a picture that proves 19 × 6 = 20 × 6 − 6. Not words — a rectangle.

C2. 998 × 5 in your head. Write the answer and the move.

....

C3. 16 × 25. Double one factor and halve the other — twice. What easy problem do you land on?

....

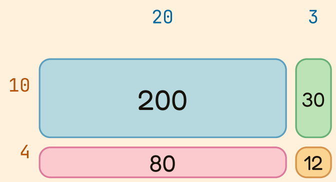
NAME ___________

DATE ___________

The Four Rooms

Two 2-digit numbers make four rooms. Every room gets a label and a number.

WORKED · 23 × 14



200+80+30+12 = 322

AXIOM · TODAY'S BRIEF

"Four rooms, always. Not three, not five. The two big rooms are easy to remember and the two small ones are where every mistake in this unit lives. Count the rooms before you add."

Tomorrow you'll go looking for rectangles nobody told you about. Bring graph paper.

✦ Warm-Up

TENS TIMES TENS

10 × 10	20 × 30	40 × 50	12 × 10	60 × 70	90 × 80
______	______	______	______	______	______

◆ Core

DRAW FOUR ROOMS. LABEL ALL FOUR. THEN ADD.

1. 23 × 14 = ______

2. 36 × 25 = ______

3. 47 × 32 = ______

4. 58 × 46 = ______

5. 64 × 27 = ______ List the four rooms in order, biggest first: ....

★ Challenge

C1. Do 23 × 14 with the standard algorithm. The first line is 92. Which *two* rooms of the area model add up to 92?

....

C2. Find two 2-digit numbers whose product is exactly 1000. There is more than one pair.

....

C3. 25 × 25 = 625. Use that to get 26 × 26 *without* multiplying it out. Explain the rooms you added.

....

Rectangle Hunt

No Warm-Up. No Core. Today is all Challenge tier — and it's the best day of the week.

Hunt 1 · Every rectangle

On graph paper, draw every rectangle with whole-number sides for each area. Count them.

AREA 36 → ______ rectangles

AREA 48 → ______ rectangles

AREA 37 → ______ rectangles

AREA 100 → ______ rectangles

Hunt 2 · The strange ones

Check every number from 1 to 30. Sort them.

EXACTLY 1 RECTANGLE:

EXACTLY 2 RECTANGLES:

AN *ODD* NUMBER OF RECTANGLES:

The real question

Some numbers have an odd number of rectangles. All the others have an even number. Find the pattern, then explain *why* it happens. This is a proof, not a guess — a mathematician would want your reason, not your list.

Missing-digit puzzles

P1. $6 \times 4\square = 276$

P2. $2\square3 \times 4 = 932$

P3. $1\square \times 15 = 240$

P4. $3\square \times 7 = 252$

P5. $\square\square \times 11 = 484$

No trial and error. For each one, write the clue you used first.

Game · Product Blockout

You need: two dice, a 10 × 10 grid, two colours.

1. Roll both dice. Multiply them.
2. Shade any rectangle on the grid with that area — your choice of shape.
3. Rectangles may not overlap.
4. If nothing fits, you pass.
5. Last player to place a rectangle wins.

Playing 6 × 6 as 4 × 9 is the whole game. Save the awkward shapes for late.

AXIOM: The odd-number question has stopped grown-ups. If you crack it, you've done real mathematics this week — write it in the journal even though you got it right.

Weeks 2–5 at a Glance

Same three-tier shape as Week 1, every day. Objectives and problem sets outlined here; full worksheets follow the Week 1 pattern.

Week 2 · The Algorithm

THE BIG QUESTION GETS ANSWERED THIS WEEK

2.1 Mon
Lay the algorithm over the model. Match each line to a room.

2.2 Tue
Algorithm fluency with carrying; area model as the checker.

2.3 Wed
3-digit × 1-digit. Same logic, longer rectangle.

2.4 Thu
Estimate first, then compute. Catch your own errors.

Fri
Lattice multiplication detour. Is it better? Defend an answer.

Week 3 · Factors

MID-UNIT QUIZ FRIDAY · 85% TO ADVANCE

3.1 Mon
Factor pairs as rectangles. Every rectangle is a pair.

3.2 Tue
Factor rainbows. Why do the arcs stop in the middle?

3.3 Wed
Common factors of two numbers; greatest common factor.

3.4 Thu
Square numbers: why do they have an odd number of factors?

Fri
Mid-unit quiz, 8 items, Weeks 1–3.

Week 4 · Multiples & Primes

NEVER SKIP THE SIEVE

4.1 Mon
Multiples on a 100-grid. Find where two patterns overlap.

4.2 Tue
Divisibility tests for 2, 3, 5, 9, 10 — and why the 3-rule works.

4.3 Wed
Sieve of Eratosthenes. Every prime under 100 on one page.

4.4 Thu
Prime factor trees. Break a number down to primes.

Fri
Twin primes hunt. Nobody knows if they ever run out.

Week 5 · Proof & Project

UNIT TEST FRIDAY · BADGE AWARDED

5.1 Mon
Missing-digit puzzles. Reason backwards from the product.

5.2 Tue
Launch Bay Blueprint: design the hangar, compute area two ways.

5.3 Wed
Blueprint defence. Prove both methods agree, out loud.

Thu
Error journal sweep. Fix only what repeats.

Fri
Mission 01 test: 12 items + one explanation.

MATERIALS, WHOLE UNIT

Base-ten blocks · two dice · deck of cards · graph paper (¼ in) · coloured pencils · one spiral notebook for the error journal

WEEKLY CHECK

Five items every Friday, mixed tiers. Below 4/5 means Monday is a reteach, not new material.

IF HE'S ABOVE 90% ON CORE

The tier is too easy. Move Challenge into the Core slot and keep going. Do not add more problems at the same level.

Mid-Unit Quiz

8 items · Weeks 1–3 · 7 of 8 keeps you flying

NAME ___________

DATE ___________

___ / 8

1. $7 \times 8 =$ ....

2. Draw 6×24 as an area model. Label both rooms and write the product.

3. 34×12. Any method you like — but show the method.

4. 19×8 in your head. Answer, then the move.

5. Every factor pair of 36.

6. Is 51 prime? Prove your answer.

7. A rug is 15 ft by 12 ft. What is its area?

CHALLENGE ITEM

8. $4\square \times 6 = 258$ $\square =$ _____

Clue you used first: ....

Mission 01 Test

12 items + one explanation · part 1 of 2

NAME ___________

DATE ___________

1. $9 \times 7 =$ ....

2. $12 \times 6 =$ ....

3. 8 × 43 with an area model. = ________

4. 7 × 86. = ________

5. 26 × 14, four rooms labelled. = ________

6. 45 × 32, any method. = ________

7. 49 × 6 in your head. Write the answer, then name the strategy.

8. Every factor pair of 48.

9. The first five multiples of 8.

Reasoning & the Big Question

ITEMS ___ / 12
EXPLANATION ___ / 4

10. Which of 2, 3, 5, 9 divide 234? Show the test you used for each.

11. Prime or composite? 61 ______ 57 ______.
How do you know?

CHALLENGE ITEM

12. 3☐ × 29 = 986 ☐ = _____

First clue:

EXPLAIN YOUR THINKING · WORTH 4 POINTS

Why does the area model give the same answer as the standard algorithm?

Draw 23 × 14 both ways. Point to where each of your four rooms shows up inside the algorithm. Then answer in your own words. Spelling doesn't count. Reasoning does.

TROPHY BAND · CIRCLE ONE

85–100%
Mission Commander
Rectangle Badge earned. Mission 02 opens — and Challenge becomes your Core.

70–84%
Flight Engineer
Two targeted days on the flagged items, then retake those items only.

Below 70%
Ground Crew
Rebuild Week 1 with physical blocks before returning to the algorithm.

Key A · Pre-assessment & Lessons 1.1–1.2

Pre-assessment

A1 56 · A2 54

B1 4×13 = 40 + 12 = 52

B2 6×47 = 240 + 42 = 282

C1 23×14 = 200+80+30+12 = 322

C2 47×32 = 1200+80+210+14 = 1504

D1 1×24, 2×12, 3×8, 4×6

D2 No — 51 = 3 × 17

E1 7, 14, 21, 28, 35

E2 2, 3, 5, 9 all divide 90

F1 114 (20×6 - 6)

F2 ▢ = 6 (6 × 46 = 276)

Scoring note. B and C are correct only if the partial products are labelled. A right answer with no visible decomposition scores 0 for those items — the model is the skill being assessed, not the product.

Lesson 1.1 · Arrays Become Area

Warm-Up 42 · 32 · 54 · 49 · 36 · 55

Core

1. 4×13 = 40+12 = **52**
2. 6×14 = 60+24 = **84**
3. 3×27 = 60+21 = **81**
4. 5×18 = 50+40 = **90**
5. 7×16 = 70+42 = **112**

Challenge. C1 · 14 (84 ÷ 6). C2 · five rectangles: 1×36, 2×18, 3×12, 4×9, 6×6 — complete because you only need to test up to 6, since 6×6 = 36 and pairs repeat after that. C3 · only 1×37, because 37 is prime; a prime has exactly one factor pair.

Lesson 1.2 · Breaking Apart

Warm-Up 45 · 64 · 56 · 54 · 48 · 33

Core

1. 240 + 32 = **272**
2. 240 + 42 = **282**
3. 180 + 54 = **234**
4. 350 + 56 = **406**
5. 360 + 24 = **384**

Challenge. C1 · accept any valid split, e.g. 8×17 = 136, then 136×2 = 272; or 8×35 − 8 = 272. C2 · ▢ = 6, ▲ = 7 (6 × 46 = 276). Strong reasoning: the product ends in 6, and 6 × ▢ must end in 6, so ▢ is 1 or 6; 41 is too small, so 46. C3 · Yes, both 272 — doubling one factor and halving the other leaves the product unchanged (the rectangle is cut and re-stacked).

TEACHER PAGE · ANSWER KEY

Key B · Lessons 1.3–1.4, Enrichment, Quiz

Lesson 1.3 · Mental Math Moves

Warm-Up 100 · 200 · 300 · 140 · 60 · 270

Core (answer · move)

1. **114** · 20×6 - 6
2. **693** · 100×7 - 7
3. **240** · half of 48×10
4. **300** · 25×4 = 100, ×3
5. **408** · 100×4 + 2×4
6. **210** · 35×6 = 30×6 + 5×6

Challenge. C1 · a 20-by-6 rectangle with a 1-by-6 strip cut off the end; the strip is the 6 he subtracts. C2 · 4990 (1000×5 – 2×5). C3 · 16×25 → 8×50 → 4×100 = **400**.

Lesson 1.4 · The Four Rooms

Warm-Up 100 · 600 · 2000 · 120 · 4200 · 7200

1. 200+80+30+12 = **322**
2. 600+150+120+30 = **900**
3. 1200+210+80+14 = **1504**
4. 2000+320+300+48 = **2668**
5. 1200+420+80+28 = **1728**

Challenge. C1 · 20×4 = 80 and 3×4 = 12; together 92, which is the algorithm's first line. C2 · 20×50 and 25×40. C3 · 676. Going from 25×25 to 26×26 adds a 1×25 strip, another 25×1 strip, and one 1×1 corner: 625 + 25 + 25 + 1 = 676.

Friday · Rectangle Hunt

Hunt 1. Area 36 → 5 (1×36, 2×18, 3×12, 4×9, 6×6). Area 48 → 5 (1×48, 2×24, 3×16, 4×12, 6×8). Area 37 → 1. Area 100 → 5 (1×100, 2×50, 4×25, 5×20, 10×10).

Hunt 2. Exactly one rectangle: 1 and the primes 2, 3, 5, 7, 11, 13, 17, 19, 23, 29. Exactly two: 6, 8, 10, 14, 15, 21, 22, 26, 27. Odd number of rectangles: 1, 4, 9, 16, 25 — the perfect squares.

The real question. Rectangles come in pairs, because every factor has a partner (a 3×12 is also a 12×3, so you only draw one of them). The count is odd exactly when one factor is its own partner — when the number is a perfect square and the rectangle is a square. Accept any answer that gets to "the square pairs with itself," in any wording.

Puzzles P1 □=6 · P2 □=3 · P3 □=6
P4 □=6 · P5 □□=44

Mid-Unit Quiz · out of 8

1. **56**
2. 6×24 = 120 + 24 = **144**
3. 34×12 = 300+60+40+8 = **408**
4. **152** · 20×8 - 8
5. 1×36, 2×18, 3×12, 4×9, 6×6
6. Not prime — 51 = 3 × 17
7. **180 sq ft**
8. □ = 3 (43 × 6 = 258)

7/8 or better → continue. Below that, reteach only the items missed. A miss on 2 or 3 means the area model needs blocks again before Week 4.

Key C · Mission 01 Test & Scoring

Items 1–12 · one point each

1. **63**
2. **72**
3. 8×43 = 320 + 24 = **344**
4. 7×86 = 560 + 42 = **602**
5. 26×14 = 200+80+60+24 = **364**
6. 45×32 = 1200+150+80+10 = **1440**
7. **294** · 50×6 - 6
8. 1×48, 2×24, 3×16, 4×12, 6×8
9. 8, 16, 24, 32, 40
10. 2 (even), 3 (digits sum 9), 9 (digits sum 9). Not 5.
11. 61 prime; 57 composite (3 × 19)
12. ▢ = 4 (34 × 29 = 986)

Explanation · 4 points

Score the reasoning, not the writing. Award one point each:

- ☐ 1 · Both methods drawn/worked, answer 322
- ☐ 2 · All four rooms labelled (200, 80, 30, 12)
- ☐ 3 · Links 92 to the 80 + 12 rooms and 230 to the 200 + 30 rooms
- ☐ 4 · States that both methods add the same four pieces, in a different order

Model answer, kid-sized. "The algorithm hides the rooms. 92 is the two rooms next to the 4, and 230 is the two rooms next to the 10. It's the same four pieces, just added in a different order."

Total 16 points. 14+ Mission Commander · 11–13 Flight Engineer · 10 or fewer Ground Crew. Item 12 missed alone never blocks the badge.

Error Journal · photocopy this page

One row per mistake, for the whole mission. The "why" line must name a step. "I was careless" doesn't count.

DATE	PROBLEM	I GOT / ANSWER	WHY IT HAPPENED — ONE SENTENCE